MIL-M-38789A
15 October 1970
SUPERSEDING
(See 6.2)

MILITARY SPECIFICATION

MANUALS, TECHNICAL: OVERHAUL INSTRUCTIONS AND OVERHAUL INSTRUCTIONS WITH ILLUSTRATED PARTS BREAKDOWN

(For Various Equipment and Accessories)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 *Coverage.* This specification prescribes the requirements for the preparation of technical manuals covering the overhaul of complex and non-complex types of equipment, such as accessories; components of aircraft; electrical equipment; electronic equipment; instruments; photographic equipment; ground support equipment (GSE); field and shop equipment; and vehicles. An overhaul instruction manual will usually be required for a complex equipment item and an overhaul instructions with illustrated parts breakdown manual will usually be required for non-complex equipment; however, the type of manual to be prepared for any particular item of equipment will be determined by the procuring activity.

1.1.1 *Overhaul instructions manual.* This type of manual shall be prepared when the complexity of the equipment will not permit complete and adequate overhaul instructions and parts breakdown information to be furnished in a combined manual prepared in accordance with the requirements and limitations prescribed herein. (Complete and adequate information is essential; instructions shall not be limited for the sole purpose of meeting the requirements for a combined overhaul instructions with illustrated parts breakdown manual, in lieu of an overhaul instructions manual and a separate illustrated parts breakdown manual.)

1.1.2 *Overhaul instructions with illustrated parts breakdown manual.* This type of manual shall be prepared when the equipment is relatively simple (non-complex), and complete and adequate overhaul instructions and parts information can be furnished in this type of manual prepared in accordance with the requirements and limitations prescribed herein. This type of manual shall not exceed 36 pages, unless contractually authorized otherwise by the procuring activity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal; form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military	
MIL-M-8910	Manuals, Technical: Illustrated Parts Breakdown; Preparation of
MIL-M-38780	Manual, Technical: Nondestructive Inspection
MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

STANDARDS

Military	
MIL-STD-681	Identification Coding for Application of Hookup and Lead Wire
MIL-STD-838	Lubrication of Military Equipment

HANDBOOK

Military	
MIL-HDBK-275	Guide for the Use of Lubricants, Compounds, and Fluids in Aircraft

PUBLICATIONS

Technical Order	
1-1-2	Corrosion Control and Treatment for Aerospace Equipment
12-1-3	Climatic Deterioration Prevention Treatment, Electronic Test and Communications Equipment

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 *General.* The general manner of preparation shall be in accordance with MIL-M-38784. Every effort shall be made to include only unclassified information in the manual. In the event that insertion of classified information is essential, the provisions of MIL-M-38784 shall apply. A manual prepared for single-point-repair shall contain the minimum essential data for experienced maintenance specialists.

3.1.1 *Text.* The requirements of MIL-M-38784 are applicable.

3.1.1.1 *Special processes.* The manual shall include information on any special procedures or processes required under extreme temperature and humidity conditions within the limits established by the design specification.

3.1.1.2 *Reference to components.* Reference to components of the equipment shall be by type designation, standard nomenclature, location reference designation, or figure and index number, as applicable.

3.1.1.3 *Color coding.* Reference to color coding of chassis wiring shall normally be in accordance with MIL-STD-681. Chassis wiring in equipment which does not conform to the requirements of MIL-STD-681 shall be explained in the manual.

3.1.1.4 *Lubrication.* The requirements of MIL-STD-838 and MIL-HDBK-275 are applicable.

3.1.2 *Illustrations.* The requirements of MIL-M-38784 are applicable.

3.1.2.1 *Diagrams.* Diagrams of systems, such as fuel, oil and hydraulic, and internal wiring diagrams (schematics) shall be furnished when necessary and pertinent to the work described. When the circuitry is electronic rather than electrical, the schematics and other wiring diagrams shall show a reference symbol for each component included in the diagrams. In such cases, a table shall also be shown on, or adjacent to, the diagrams listing the components by basic noun nomenclature and model designation, and giving the blocks of reference symbol numbers assigned to each component.

3.1.2.1.1 A table shall be provided and shall show the nominal current, voltage, or frequency, and the voltage or current wave shape associated with each test point during normal operation.

3.1.2.2 *Exploded views.* Exploded views of the equipment shall be used for reference in disassembly and assembly instructions. Index numbers and a nomenclature list or legend, keyed to the index numbers, shall be used to identify each part. If the equipment is of such a nature that it cannot be adequately illustrated by a single exploded view, it may be exploded by subassemblies as separate views. In such cases, an exploded view showing the complete equipment exploded into its major assemblies shall be shown first. Parts which attach, and connect, the major assemblies together shall be shown on this illustration. These views, and those in a separate illustrated parts breakdown shall be the same, with the sequence of key index numbers in the order of disassembly.

3.1.2.3 *Procedural illustrations.* Essential illustrations depicting mechanical operations shall be included in pertinent sections, as necessary. It is not necessary to illustrate each step of an overhaul procedure, such as the removal of screws with an ordinary screwdriver, lifting off a cover after the screws have been removed, etc. Procedural illustrations should definitely supplement the text by clarifying procedures which are of a special nature or are not obvious.

3.1.2.4 *Tool and test equipment illustrations.* Standard tools and test equipment shall not be illustrated as such, nor shall self-evident or generally known uses be shown. Only uncommon or unusual uses, and connections for test purposes need be illustrated in appropriate sections if it is essential to do so to avoid misunderstanding as to proper procedures. Unusual operations in checking fits, clearances, backlash, etc, with feeler gages, torque wrenches, and micrometers shall be illustrated.

3.1.3 *Test point identification symbols.* A system of test point identification by symbols shall be incorporated on all schematic wiring diagrams, tables, charts, and illustrations. The following system is commonly used. A contractor's system is acceptable. The introduction shall explain the system.

3.1.3.1 *Major test points.* A star-encircled Arabic numeral shall be used to identify and designate the test points used in checking the over-all functions of, and localizing trouble to, (a) individual components, and (b) assemblies, such as i-f strips, oscillator circuits, or amplifier stages, (c) groups of components. They will always be identified on schematic wiring diagrams, tables, charts, and illustrations by use of an Arabic numerical, 1, 2, 3, etc, enclosed in a star, but will be referred to in the text as test point 1, test point 2, etc. Typical examples of major test points are: (a) the input terminals for the supply source voltage; (b) the high voltage output terminals of the equipment's power system; (c) the signal input terminals of the equipment; (d) the signal output terminals of the equipment; (e) input and output points of an i-f or a-f amplifier; (f) oscillator circuits; and (g) the major points of power or voltage distribution within the equipment.

3.1.3.2 *Secondary test points.* An encircled capital letter Ⓐ shall be used to identify and designate

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the test points used in isolating causes of subnormal performance within a specific assembly or subassembly of the equipment. They will always be identified on schematic wiring diagrams, tables, charts, and illustrations by use of capital letters, A, B, C, through Z, AA, AB through AZ, BA, BB through BZ, etc, enclosed in a circle, but will be referred to in the text as test point A, test point B, etc. Typical examples of secondary test points are: (a) circuit supply voltage terminals; (b) signal injection points for gain measurements of a stage or a group of stages; (c) signal output points used in checking the gain of a stage or a group of stages.

3.1.3.3 *Minor test points.* An encircled capital letter and Arabic numerical (both in a single circle) (A1) shall be used to identify and designate the test points used in isolating causes of abnormal indication within a specific circuit of the equipment. They will always be identified on schematic wiring diagrams, tables, charts, and illustrations by use of a capital letter and Arabic numeral, A1, A2, etc, enclosed in a circle but will be referred to in the text as test point A1, test point A2, etc. Typical examples of minor test points to be so designated are: (a) signal injection and output points of a stage for a check or measurement of gain; (b) specific voltage and resistance check point; (c) all other points used in checking for trouble in any specific circuit. Different letters shall be assigned each component.

3.1.3.4 *Restricted usage.* Indiscriminate use of major and secondary test points shall be avoided; for example, test point identification should normally be restricted to signal tracing test points and to voltage and continuity check points.

3.1.3.5 *Assignment of symbols.* Test point identification symbols on the schematic wiring diagrams shall be assigned consecutively throughout the composite equipment. Once assigned, a symbol shall not be reassigned to another test point. An explanation of the use of test points shall appear in Section I.

3.2 Arrangement. Manuals shall be arranged in the following manner. Variation in sectionalization is permissable for certain types of manuals, dependent upon complexity of equipment. For noncomplex items, material for two or more sections may be combined and some sections, not required, may be omitted. Section numbers and titles shall be changed accordingly. Manuals on simple items, usually eight pages or less, may be divided into paragraphs instead of sections.

Front Matter
- Title Page
- List of Effective Pages
- Promulgation Page
- Tables of Contents
- List of Illustrations
- List of Tables

Introduction

Section I	General Information
Section II	Special Tools and Test Equipment
Section III	Disassembly
Section IV	Cleaning
Section V	Inspection, Repair, and Replacement
Section VI	Assembly
Section VII	Accessories
Section VIII	Testing
Section IX	Table of Limits
Section X	Illustrated Parts Breakdown
Section XI	Difference Data Sheets

3.2.1 *Front matter.*

3.2.1.1 Front matter shall be prepared in accordance with MIL-M-38784.

3.2.1.1.1 The introduction shall provide a brief explanation of the manual's purpose, scope, and arrangement including any relevant information which may increase the usability of the manual. The item(s) covered shall be identified by government designation and manufacturer's type or model designation. An illustration or exploded view shall be included.

3.2.1.1.2 If the manual will include an Illustrated Parts Breakdown and repair parts for the equipment are to be supplied in the form of kits, the Introduction of the manual shall contain the following statement: "Some repair parts for the equipment covered by this manual are provided in the form of kits. The Illustrated Parts breakdown section hereof identifies these parts. Activities shall replace all parts, regardless of condition, which are removed in the process of disassembly with all like parts furnished in the kit. Therefore, instructions for cleaning, inspecting, and reworking such used parts have been omitted. If any parts in the kit must be cleaned, inspected, or tested prior to installation, instructions for performing these requirements are included in this manual. Naturally, all defective parts are to be replaced; but a part unnecessary to be removed in the process of disassembly, shall not be removed solely for the purpose of replacement by a corresponding kitted part. For Air Force, refer to AFLCR 65-42 regarding residue from kits and removed parts in this category."

3.2.1.1.3 If applicable, environmental control (clean room) information required for overhaul of the equipment shall be specified.

3.2.2 *Section I, General Information.*

3.2.2.1 This section shall provide general information to be used by maintenance personnel to further their understanding of the equipment and its related systems. A brief description of the equipment shall be provided reflecting purpose, type, series, a table of leading particulars, and main features.

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3.2.2.2 If the manual covers more than one model, significant differences shall be explained. If "Difference Data Sheets" are contained in the manual, the introduction need only refer to them for explanation of differences.

3.2.3 *Section II, Special Tools and Test Equipment.* This section shall list and illustrate special tools and test equipment required for the disassembly, inspection, repair, assembly, and testing of the components and end item. Items shall be selected from the list of government approved tools and test equipment. When specified, the contractor's recommended tools and test equipment shall be shown. However, these recommendations shall be changed promptly, if necessary, to conform to the official government list as soon as such information is available. Acceptable alternate tools and test equipment will also be listed. Standard types of tools such as screw drivers, pliers, soldering irons, shall not be listed. Standard types of test equipment such as voltmeters, signal generators, tube testers, shall be listed.

3.2.3.1 *Special tools and test equipment list.* Special tools and test equipment shall be listed in the following format:

SPECIAL TOOLS AND TEST EQUIPMENT LIST

Tool/Equipment Number	*Figure No.*	*Nomenclature*	*Use and Application*

a. Tool/Equipment Number—Tools shall be listed in sequence by part (tool) number. The part (tool) number shall agree with the drawing number. Test equipment shall follow the tool list and be listed in alpha-numeric sequence by type designation, using the AN type designation if assigned, or the commercial or manufacturer's designation if an AN type was not assigned. If an alternate item can be used, it will be listed immediately following the prime item and be identified as an alternate for the preceding item.

b. Figure No.—This column shall contain the number of the applicable illustration in which the tool/equipment is shown.

c. Nomenclature—The nomenclature shall agree with the drawing.

d. Use and Application—The purpose of each item shall be stated. Reference to the applicable paragraph in the text will suffice. If the manual covers more than one model, application to each model shall be indicated.

3.2.3.2 *Illustrations of special tools and test equipment.* Special tools and test equipment shall be illustrated in this section unless they are adequately illustrated in other sections of the manual. Standard items need not be illustrated.

3.2.3.3 *Test equipment hook-up.* If applicable, instructions shall be included for fabricating interconnecting cable assemblies, etc, required for bench test set-ups and not normally supplied with the equipment.

3.2.4 *Section III, Disassembly.* This section shall present, by text and illustrations, any special procedures preliminary to inspection, repair, and replacement of parts. These special procedures consist of removal from shipping container, dismantling into major assemblies, subassemblies, and component parts. The text shall contain statements keyed to the index numbers of the illustration, explaining special precautions to be observed and mentioning the special tools to be used to remove or replace certain parts or assemblies. Critical dimensions to be taken during disassembly should be specific.

3.2.4.1 *Inspect, Repair, As Necessary.* Data shall be included to enable depot activities to determine when components, assemblies, and subassemblies require maintenance or repair, or when they shall be completely overhauled. Processing instructions shall be included to cover inspection, the extent of disassembly necessary and the detailed mandatory parts replacement information applicable to time replacement and forced removal components.

3.2.5 *Section IV, Cleaning.* This section shall describe any special cleaning methods required. The type of cleaning shall be specified, such as using a liquid, brushing away dust, blowing out with compressed air; if the latter, air pressure required shall be indicated. Requirements for special ventilation or controlled environment should be indicated. Precautions to be observed shall be listed. Instructions for after cleaning relating to the preservation of metal parts and surface treatment (such as paint touch-up) shall be provided. Cleaning instructions in connection with nondestructive inspection methods shall be included. Special tools, jigs, fixtures, necessary during cleaning shall be indicated. General cleaning instructions shall be omitted.

3.2.6 *Section V, Inspection, Repair, and Replacement.*

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3.2.6.1 *Inspection.*

3.2.6.1.1 This portion of the section shall include inspection instructions that may be classed as special. This means inspections that may be peculiar to a certain part of the equipment and for which maintenance personnel are likely to have little or no knowledge. Inspection instructions shall encompass failures, wear, damage, corrosion, leakage, aging, burning, malfunctioning, deformation and deterioration that can be expected to occur during normal service of the unit. They shall not include inspections that would only be applicable to detection of manufacturing defects. Each subassembly or the parts thereof, shall be treated as a separate unit in any discussion of inspection procedure. Each inspection check shall be described as the sequence of repair is required.

3.2.6.1.2 Definitions, and illustrations when necessary, of descriptive terms such as "scoring" and "galling" shall be included.

3.2.6.1.3 All parts and subassemblies shall be listed that will normally require dimensional check for wear, damage, and deformation. The limiting dimensions of tolerances shall be such that inspected parts may be separated into (a) acceptable without further processing, (b) reparable, (c) rejected. The relationship of these dimensional limits to the assembly clearances specified in the Table of Limits, Section IX, shall be such that parts not classified as "rejected" may be assembled with normal adjustments, and may reasonably be expected to be within the replacement maximum limit of the Table of Limits.

3.2.6.1.4 All parts and subassemblies shall be listed that require special inspections by non-destructive methods, such as magnetic particle, ultrasonic, eddy current, x-ray, and liquid penetrants. When such inspections are necessary, the information required by that portion of MIL-M-38780 titled "Subsequent Sections" shall be provided.

3.2.6.1.5 Particular attention shall be given to all critical areas. A chart shall be used to summarize conditions for checking individual parts. This chart shall include material specifications for metals when pertinent to making repairs.

3.2.6.1.6 A concise discussion shall be given of the Table of Limits as it applies to this section. Definite information shall be given on the correct interpretation of the "Replacement Maximum" column in the Table of Limits.

3.2.6.2 *Repair and replacement.*

3.2.6.2.1 Comprehensive instructions shall be provided for necessary repair and replacement operations to restore the equipment to a completely satisfactory condition. Text will be supplemented with illustrations as necessary to describe complex procedures. Repair methods shall be specified for the correction of all deficiencies and conditions considered reparable when inspected but shall not cover assemblies replaced as units because they cannot be economically repaired or overhauled. Description of common procedures shall be omitted. Specific data shall be provided to adequately describe acceptable repairs.

3.2.6.2.2 The application and operation of special tools and test equipment shall be described as they are required in the repair and replacement operations.

3.2.6.2.3 Instructions for corrosion/fungi prevention measures are contained in TO 1-1-2 and 12-1-3. If applicable, the manual shall reference these technical orders. Such instructions that are peculiar to the equipment shall be inserted in the manual.

3.2.7 *Section VI, Assembly.* This section shall give essential instructions for the assembly of component parts into major subassemblies and final assembly of subassemblies to form a complete end item. If assembly does not have any safety features, torque values, clearances, special techniques, etc, to be observed; and reassembly is essentially the reverse of disassembly, a statement to that effect is satisfactory. Approved corrosion-preventive compounds, special lubricants, gasket pastes, shall be specified. Safetying methods and devices shall be emphasized. Methods for making all checks for clearance, backlash, and torque value shall be described, and when necessary for clarity, illustrated. Appropriate tests for operation of a subassembly prior to final assembly shall be provided, including safety precautions to be observed. Instructions for checking the end item, such as adjustments, timing, inspection, shall be included.

3.2.8 *Section VII, Accessories.* This section is optional and shall be included in manuals covering relatively complex items upon which accessories are installed and overhaul instructions for such accessories are included in the end item manual.

3.2.8.1 Separate overhaul manuals will normally be procured for major accessories. This section shall accordingly state that for overhaul instructions on the major accessories not covered in this section, reference should be made to the applicable separate manuals.

3.2.8.2 This section shall contain applicable replacement, disassembly, special cleaning, inspection, repair, assembly, and test instructions on the minor accessories.

3.2.9 *Section VIII, Testing.* This section shall contain essential instructions for aligning, adjusting, testing, and calibrating the equipment after reassembly. Test values and acceptable tolerances shall be given for each test, as applicable. Remedial action shall also be described for failures of equipment to meet the necessary tests. This may be

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the form of a troubleshooting table, if appropriate, showing trouble, probable causes, and remedy. As far as possible, the remedies shall specify adjustments which can be made without disassembling the equipment. If operating instructions are needed by overhaul personnel to test the equipment properly, such instructions shall be included to the extent necessary. Information for performing tests in screen rooms, and for proper utilization of power absorption and reflecting test equipment to prevent radiation into the atmosphere, shall be given.

3.2.10 *Section IX, Table of Limits.* This section shall consist of introductory material, the Table of Limits, a miscellaneous table, and charts showing the limit points and lubrication system. The introductory material shall include definitions of unusual terms and abbreviations; also, a statement to the effect that a minimum and maximum are set up as ideal limits and that measurements not exceeding the replacement maximum permit the part to be continued in service.

3.2.10.1 The Table of Limits shall include all clearances, backlashes, endplays, etc, arranged in numerical sequence by reference number. Reference numbers shall be so assigned that a numerical sequence for them will place the limits for particular areas of the equipment, or for subassemblies, into groups. Limits contained in the Table of Limits section shall not be stated in the text of other sections. Reference in the text to these limits shall be made by citing the reference numbers used in the tables. The table shall be a cross-reference to the chart on which the reference number is located. The table shall have the following format:

Ref No.	*Chart No.*	*Description*	*Minimum*	*Maximum*	*Replacement Maximum*

3.2.10.2 The Table of Limits shall be supported by illustrations (charts) to show clearly all points of limits. Illustrations shall not show limit values but only index or reference numbers with leaders to the point of clearance. The illustrations shall also show lubrication information. Pressure oil, return oil, scavenge oil, propellants and pump fluids shall be differentiated by cross-hatching or other drafting means or by colors, with a legend explaining the coding.

3.2.10.3 The miscellaneous table shall include spring pressures and torque limits.

3.2.11 *Section X, Illustrated Parts Breakdown.* This section shall be included only in manuals of overhaul instructions with illustrated parts breakdown for non-complex equipment. (Refer to 1.1.2)

3.2.11.1 This section shall be generally in accordance with MIL-M-8910. Introductory information such as; explanation of symbols and abbreviations, definitions of applicable source and reparability codes, vendors' names and addressees, etc, will be included to the degree necessary to assure effective use of the manual.

3.2.11.2 An exploded view illustration and Group Assembly Parts List shall be all that is required for simple items. The format requirements of MIL-M-8910, examples 3 and 3A, are applicable.

3.2.11.3 Special tools and test equipment shall not be listed in this section since they will appear in the overhaul portion of the manual.

3.2.12 *Section XI, Difference Data Sheets.* This section, when applicable, shall consist of Difference Data Sheets for additional models of the equipment. A separate Difference Data Sheet shall be prepared for each additional model to specify in the briefest practical form how overhaul and test procedures for that model differ from the procedures given for the basic model in preceding sections.

3.2.12.1 *Format.* The requirements of MIL-M-38784 are applicable. The Difference Data Sheet(s) for a particular model shall start on a right-hand page. Page numbers shall run consecutively throughout the section. They shall be added chronologically as they are issued.

3.2.12.2 *Content.* Difference Data Sheets shall contain the following, as appropriate, to compress in one or two pages, and not more than four pages, the information on differences between the added model and the basic model covered in the preceding sections.

a. An external view of the model, if different than the one shown in Section I.

b. A table of leading particulars or a table of characteristic differences similar to that contained in Section I.

c. Illustrations which are essential for clarifying differences. Illustrations in preceding sections shall not be duplicated but referenced. Reference shall not be made to illustrations on other Difference Data Sheets.

d. Brief coverage of subjects specified for Sections II through IX by appropriate statements such as "Disassembly. Same as Model No. ——— except that ———." "Inspection. Same as Model No. ———." "Lubrication. None." Sentences shall be in short telegraphic style.

e. The last portion of the sheets for each model shall consist of a complete Group Assembly Parts List, as required by 3.2.11.2, or a list of differences arranged in Group Assembly Parts List format.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-M-38784.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784.

6. NOTES

6.1 *Intended use.* The manuals prepared in accordance with this specification are intended for use by maintenance personnel for overhauling complex and non-complex equipment.

6.2 *Supersession data.* This specification supersedes MIL-M-38789 (USAF) dated 1 July 1968.

Custodian:
Air Force—16
Navy—AS

Preparing Activity:
Air Force—16

Review Activities:
Air Force—01, 10, 16
Navy—MC

User Activities:
Air Force—01, 10, 11, 19, 70, 71, 80, 82, 84
Navy—CG, YD

Project TMSS-0044

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